

Jakob Shurtleff

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shurtlej@byui.edu • 208-206-7185 • [linkedin.com/in/jakob-shurtleff](https://www.linkedin.com/in/jakob-shurtleff)

EDUCATION

Brigham Young University - Idaho | *Rexburg, ID*

B.S. in Physics | *Expected Graduation: December 2027*

* Emphasis on computational methods and machine learning application

TECHNICAL PROJECTS

Independent Computational & AI Development * Engineered object-oriented Python applications, data processing pipelines, and automation tools using Polars and Scikit-learn. * Built custom LLM utilities and multi-agent workflows to explore automated reasoning and language processing techniques.

EXPERIENCE

BYU-Idaho Physics Department | *Rexburg, ID*

Physics Lab Coordinator & Teaching Assistant | *February 2026 – Present*

* Coordinated operations and logistics for the university physics homework help center, facilitating peer instruction and academic support resources. * Served as Teaching Assistant for Physics 121 (Principles of Physics I); clarified mechanics concepts, and graded coursework[cite: 1].

BYU-Idaho Physics Department | *Rexburg, ID*

Volunteer Physics Tutor | *August 2025 – Present*

* Conducted peer group tutoring in classical mechanics and numerical modeling to help students bridge abstract mathematics with computational applications[cite: 1].

Aptive Environmental | *Vacaville, CA*

Door-to-Door Sales Representative | *May 2025 – July 2025*

* Executed direct-to-consumer sales, honing negotiation, objection handling, and pipeline management in a performance-driven environment[cite: 1].

TECHNICAL SKILLS

- **Languages & Libraries:** Python (NumPy, Polars, Scikit-learn, Matplotlib), C++, C#, SQL[cite: 1]
 - **Developer Tools:** Git, Visual Studio, Linux, REST APIs[cite: 1]
 - **Domain Focus:** Numerical simulation, machine learning deployment, process automation, data analysis[cite: 1]
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CERTIFICATIONS

- **AI Engineering and Development** — Coursera (2025)[cite: 1]